

ON QUADRATIC FORMS FROM MIXTURES OF TWO NORMAL POPULATIONS

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Summary

In this paper, the joint distributions of the quadratic forms of the type $\mathbf{x}'\mathbf{A}\mathbf{x}$ based on samples from a population that is a mixture of two normal populations are obtained. We have also considered the joint distribution of a linear and a quadratic form.

1. Introduction

Let us consider a population which is a mixture of two normal populations in the proportion $p:(1-p)$. We can represent the probability density function of such a population as

$$(1.1) \quad f(x) = p\phi_1(x;\mu_1, \sigma_1^2) + (1-p)\phi_2(x;\mu_2, \sigma_2^2)$$

where, for $j = 1, 2$,

$$\phi_j(x;\mu_j, \sigma_j^2) = \frac{1}{\sqrt{2\pi} \sigma_j} \exp \left\{ -\frac{(x-\mu_j)^2}{2\sigma_j^2} \right\}, \quad -\infty < x < \infty.$$

It is quite simple to show that the characteristic function (c.f.) of (1.1) is

$$\psi(t) = E[\exp itx] = p\psi_1(t;\mu_1, \sigma_1^2) + (1-p)\psi_2(t;\mu_2, \sigma_2^2)$$

with $\psi_j(t;\mu_j, \sigma_j^2)$ representing the c.f.

$$\exp [it\mu_j - t^2\sigma_j^2/2] \quad j = 1, 2.$$

Hyrenius (1949) considered the distribution of the familiar ratio $\sqrt{n} \bar{x}/s$ based on the sample of size n from (1.1). He has also given the expression for the p.d.f. of the ratio s_1^2/s_2^2 based on samples from (1.1). Some Bessel function representations of the p.d.f. of

*Research supported through NRC grants.

the Student ratio are also suggested. In a later paper (1950) the author generalizes the results to a mixture of k normal populations. In the present paper we will be considering, in the main, the joint distribution of (i) two quadratic forms, (ii) a linear and a quadratic form. In this respect we generalize the results of Hyrenius.

Consider a random sample of size n from the population (1.1). Let A be a matrix of known constants $\{a_{ij}; i=1,2,\dots,n; j=1,2,\dots,n\}$ and $\ell' = \{\ell_1, \ell_2, \dots, \ell_n\}$ be a row vector also of known constants. We associate with the random sample $(x_1, \dots, x_n)$ the quadratic form $\underline{x}'Ax$ and the linear form $\underline{\ell}'x$. Let us use the notation

$$Q(x_1, \dots, x_n) = \underline{x}'Ax.$$

2. Preliminaries

If $x_1, x_2, \dots, x_n$ is a random sample of size n from the population (1.1) it is helpful to obtain a particular representation for their joint p.d.f.:

$$(2.1) \quad f(x_1, \dots, x_n) = \prod_{j=1}^n [p\phi_1(x_j; \mu_1, \sigma_1^2) + (1-p)\phi_2(x_j; \mu_2, \sigma_2^2)]$$

Upon performing the multiplication involved in (2.1) we obtain

$$f(x_1, x_2, \dots, x_n) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \phi_p^{(r)}(x_1, x_2, \dots, x_n)$$

where $\sum_p$ denotes the summation over the permutations of the set $(1, 1, \dots, 1, 2, 2, \dots, 2)$

there being r 1's and $(n-r)$ 2's. The function $\phi_p^{(r)}(x_1, \dots, x_n)$ stands for the p.d.f.

$$\phi_{i_1, \dots, i_n}^{(r)}(x_1, \dots, x_n) = \frac{1}{(\sqrt{2\pi})^n \sigma_{i_1} \sigma_{i_2} \dots \sigma_{i_n}} \exp \left[-\frac{1}{2} \sum_{j=1}^n \frac{(x_j - \mu_{i_j})^2}{\sigma_{i_j}^2} \right]$$

with r of the subscripts $(i_1, \dots, i_n)$ being 1 and $(n-r)$ being equal to 2. Thus, representing by Λ_p a diagonal matrix with the elements $\sigma_{i_1}^2, \dots, \sigma_{i_n}^2$ and by $\underline{\mu}_p$ the vector $(\mu_{i_1}, \dots, \mu_{i_n})$, we can write

$$(2.2) \quad \phi_{i_1, \dots, i_n}^{(r)}(x_1, \dots, x_n) = \frac{1}{(\sqrt{2\pi})^n |\Lambda_p|^{1/2}} \exp \left[-\frac{1}{2} (\underline{x} - \underline{\mu}_p)' \Lambda_p^{-1} (\underline{x} - \underline{\mu}_p) \right]$$

Using the representation (2.2) of $f(x_1, x_2, \dots, x_n)$ we can write down the c.f. of $Q(x_1, x_2, \dots, x_n)$ as the sum of the finite series

$$(2.3) \quad \psi_Q(t) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \int_{R_n} \exp(itQ) \phi_p^{(r)}(x_1, \dots, x_n) dx_1, \dots, dx_n.$$

Remembering that $\phi_p^{(r)}(x_1, \dots, x_n)$ is a multivariate normal p.d.f. with the variance matrix Λ_p and mean vector $\underline{\mu}_p$, we have,

$$(2.4) \quad \psi_Q(t) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p |I - 2itA\Lambda_p|^{-1/2} \exp\left[-\frac{1}{2} \underline{\mu}_p' \Lambda_p^{-1} \underline{\mu}_p\right. \\ \left. + \frac{1}{2} \underline{\mu}_p' (\Lambda_p - 2it\Lambda_p A \Lambda_p)^{-1} \underline{\mu}_p\right]$$

the familiar expression for the c.f. of a q.f. in a multivariate normal distribution.

Special Cases:

(a) If A is a diagonal matrix $(\alpha_1, \dots, \alpha_n)$, then (2.4) can be written as

$$(2.5) \quad \psi(t) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \prod_{j=1}^n (1 - 2it\alpha_j \sigma_{ij}^2)^{-1/2} \exp - \frac{1}{2} \\ \left\{ \sum_{j=1}^n \left[\frac{\mu_{ij}}{\sigma_{ij}} \right]^2 \left[\frac{-2it\alpha_j \sigma_{ij}^2}{1 - 2it\alpha_j \sigma_{ij}^2} \right] \right\}$$

We note here that in (2.5) the expression cannot be further simplified.

(i) $\sigma_1^2 = \sigma_2^2 = \lambda$, in which case we have

$$\psi(t) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \left\{ \prod_{j=1}^n (1 - 2it\alpha_j \lambda)^{-1/2} \exp - \frac{1}{2} \sum_j \left[\frac{\mu_{ij}^2 (-2it\alpha_j)}{(1 - 2it\lambda\alpha_j)} \right] \right\}$$

(ii) If, on the other hand, $\alpha_j = 1$, we have

$$Q = \sum_{i=1}^n x_i^2$$

then

$$\psi(t) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \left\{ \prod_{j=1}^n (1 - 2it\sigma_{ij}^2)^{-1/2} \right\} \exp \left\{ -\frac{1}{2} \sum_{j=1}^n \left[\frac{\mu_{ij}^2 (-2it)}{1 - 2it\sigma_{ij}^2} \right] \right\}$$

This may be simplified by remembering that the product and the summation can be written as

$$(1 - 2it\sigma_1^2)^{-r/2} (1 - 2it\sigma_2^2)^{-(n-r)/2}$$

and

$$2it \left[\frac{r\mu_1^2}{1-2it\sigma_1^2} + \frac{(n-r)\mu_2^2}{1-2it\sigma_2^2} \right]$$

Hence

$$\psi(t) = \left[p \frac{\exp \{it\mu_1^2/(1-2it\sigma_1^2)\}}{(1-2it\sigma_1^2)^{1/2}} + (1-p) \frac{\exp \{it\mu_2^2/(1-2it\sigma_2^2)\}}{(1-2it\sigma_2^2)^{1/2}} \right]^n$$

as could be obtained directly.

(b) If $A = (I-J/n)$, where I is the identity matrix and J is the matrix of 1's everywhere, then:

$$Q = \underline{x}'\underline{x} - \frac{1}{n} \underline{x}'J\underline{x} = \sum_{i=1}^n (x_i - \bar{x})^2.$$

If $\sigma_1^2 = \sigma_2^2 = \lambda$, then

$$|I-2it\lambda A_p| = |I-2it\lambda(I-J/n)|$$

which can be shown to be equal to $(1-2it\lambda)^{(n-1)}$. (See, for example, Rao, 1965). Also one has

$$\{A_p - 2it\lambda A_p A_p\}^{-1} = \{\lambda I - 2it\lambda^2(I-J/n)\}^{-1}.$$

Since $J = \underline{j}\underline{j}'$, therefore, the matrix has the inverse, Rao (1965),

$$\{I - 2it\lambda \frac{J}{n}\}^{-1} = \lambda(1-2it\lambda).$$

Thus, the c.f. of $\Sigma(x_i - \bar{x})^2$ can be shown to be

$$\begin{aligned} \psi(t) = & \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p (1-2it\lambda)^{-(n-1)/2} \exp \left[- \left(\sum_{j=1}^n \mu_{ij}^2 \right) / 2\lambda \right. \\ & \left. + \left\{ \sum_{j=1}^n \mu_{ij}^2 - 2it\lambda(\Sigma \mu_{ij})^2/n \right\} / 2\lambda(1-2it\lambda) \right] \end{aligned}$$

which is simplified to

$$(2.6) \quad \psi(t) = \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} \exp \left\{ -\frac{\delta_r^2}{2\lambda} \left(\frac{-2it\lambda}{1-2it\lambda} \right)^{n-1} \right\} (1-2it\lambda)^{(n-1)/2}$$

upon using the identities

$$(2.7) \quad \begin{cases} \sum_{j=1}^n \mu_{ij}^2 = r\mu_1^2 + (n-r)\mu_2^2 \\ \sum_{j=1}^n \mu_{ij} = r\mu_1 + (n-r)\mu_2 \\ \text{and} \\ [r\mu_1^2 + (n-r)\mu_2^2] - [r\mu_1 + (n-r)\mu_2]^2/n = \frac{r(n-r)}{n} (\mu_1 - \mu_2)^2. \end{cases}$$

In equation (2.6) we have represented by δ_r^2 the quantity $[r(n-r)(\mu_1 - \mu_2)^2/n]$. One notes here that the summand in the expression for $\psi(t)$ is a weighted characteristic function of a non-central chi-square variable times λ with the non-centrality parameter $\delta_r^2/2\lambda$ and $\nu = (n-1)$ degrees of freedom. The weights are the binomial coefficients. Hence the p.d.f. of $\Sigma(x_{ij} - \bar{x}_i)^2$ is given by

$$(2.8) \quad g(s^2; \lambda, n) = \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} f(\chi_{n-1}^2; \delta_r^2/2\lambda).$$

Referring to Plackett (1960), one obtains the Bessel function representation for the p.d.f. as

$$(2.9) \quad g(s^2; \lambda, n) = (2\lambda)^{-1} \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} \left(\frac{s}{\delta_r} \right)^{\frac{n-3}{2}} \exp \left\{ -\left(\frac{s^2 + \delta_r^2}{2} \right) \right\} I_{(n-3)/2}(s\delta_r/\lambda),$$

$s^2 > 0$

with $I_j(x)$ representing the complex integral of a Bessel function with an imaginary argument.

- (c) One way classification: Let $\{x_{ij}; i=1,2,\dots,k; j=1,2,\dots,n\}$ be a random sample of size nk from a population of the type (1.1). Let us define the residual or within sum of squares as

$$W = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2.$$

This quantity can be expressed in matrix notation as

$$W = \underline{x}' [I - \Gamma/n] \underline{x}$$

where I is an $(nk \times nk)$ identity matrix, and

$$(2.10) \quad \Gamma = \text{diag}\{J, J, \dots, J\}$$

i.e., a matrix partitioned into $k \times k$ sub-matrices with the matrix $J(n \times n)$ along the main diagonal and the zero-matrix elsewhere. In this case we have for the c.f. of W (writing $N = nk$)

$$\begin{aligned} \psi(t) = & \sum_{r=0}^N p^r (1-p)^{N-r} \sum_p [|I - 2it(I - \Gamma/n)\Lambda_p|^{-1/2}] \\ & \cdot \exp \left\{ -\frac{1}{2} \left\{ \underline{\mu}'_p \Lambda_p^{-1} \underline{\mu}_p + \underline{\mu}'_p [\Lambda_p - 2it\Lambda_p(I - \Gamma/n)\Lambda_p]^{-1} \underline{\mu}_p \right\} \right\}. \end{aligned}$$

Let us assume here that $\lambda = \sigma_1^2 = \sigma_2^2$; then the determinant

$$|I - 2it\lambda(I - \Gamma/n)| = (1 - 2it\lambda)^{(N-k)}$$

and the matrix

$$[\Lambda_p - 2it\Lambda_p(I - \Gamma/n)\Lambda_p]^{-1}$$

is equal to

$$[I - 2it\lambda(I - \Gamma/n)]^{-1}/\lambda.$$

But

$$[I(1 - 2it\lambda) + (2it\lambda/n)\Gamma]^{-1} = (1 - 2it\lambda)^{-1} \left\{ I + \left[\frac{2it\lambda}{1 - 2it\lambda} \right] \frac{\Gamma}{n} \right\}^{-1}$$

and

$$I + \{2it\lambda/n(1 - 2it\lambda)\}\Gamma/n = \text{diag} \{I + cJ, \dots, I + cJ\}$$

and hence has the inverse with

$$(I + cJ)^{-1} = I - \frac{2it\lambda}{n} J$$

along its diagonal and zero-matrix elsewhere. Thus

$$\begin{aligned} & \mu_p' [\Lambda_p - 2it\Lambda_p(I - \Gamma/n)\Lambda_p]^{-1} \mu_p \\ &= \{\lambda(1-2it\lambda)\}^{-1} \left\{ \sum_{j=1}^{nk} \mu_{i_j}^2 - \frac{2it\lambda}{n} \sum_{h=1}^k \left[\sum_{j=1+h^*}^{n+h^*} \mu_{i_j} \right]^2 \right\} \end{aligned}$$

with $h^* = n(h-1)$. Therefore, the characteristic function of W can be simplified to the form

$$(2.11) \quad \psi(t) = \sum_{r=0}^N p^r (1-p)^{N-r} \sum_p (1-2it\lambda)^{-(N-k)/2}$$

$$\exp \left\{ \frac{it\lambda}{(1-2it\lambda)} \sum_h \sum_j \left[\frac{(\mu_{i_j} - \bar{\mu}_{h^*})^2}{\lambda} \right] \right\},$$

where

$$\bar{\mu}_{h^*} = \frac{1}{n} \sum_{j=1+h^*}^{n+h^*} \mu_{i_j}.$$

This is again a weighted sum of the characteristic function of $\lambda\chi^2_{(N-k)} \delta_p^2$ with

$$(2.12) \quad \delta_p^2 = \sum_h \sum_j (\mu_{i_j} - \bar{\mu}_{h^*})^2 / 2\lambda$$

and depends upon the particular permutation of the μ_1 's and μ_2 's. If $\mu_1 = \mu_2$ (2.9) reduces to the normal form

$$(1-2it\lambda)^{-(N-k)/2}.$$

3. Joint distribution of quadratic forms

3.1 Consider the quadratic forms $\underline{x}'A\underline{x}$ and $\underline{x}'B\underline{x}$ with the joint characteristic function

$$\psi(t_1, t_2) = E[\exp \{it_1 \underline{x}'A\underline{x} + it_2 \underline{x}'B\underline{x}\}].$$

Upon writing

$$C = At_1 + Bt_2$$

we see, at once, that

$$\psi(t_1, t_2) = E[\exp i \{ \underline{x}' C \underline{x} \}]$$

Thus the characteristic function $\psi(t_1, t_2)$ can be determined from (2.4) by putting $t=1$ and replacing A by C viz.,

$$(3.1) \quad \psi(t_1, t_2) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \{ |I - 2i(t_1 A + t_2 B) \Lambda_p|^{-1/2} \exp[-(\mu'_p \Lambda_p^{-1} \mu_p)/2 + \mu'_p \{ \Lambda_p^{-2i} (\Lambda_p A t_1 \Lambda_p + \Lambda_p B t_2 \Lambda_p) \}^{-1} \mu_p / 2] \}.$$

3.2 Special cases

(a) If A and B are diagonal $\{\alpha_1, \dots, \alpha_n\}$ $\{\beta_1, \beta_2, \dots, \beta_n\}$ respectively, then the equation (3.1) reduces to

$$(3.2) \quad \psi(t_1, t_2) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p \left[\prod_{j=1}^n [1 - 2i\sigma_{1j}^2 (t_1 \alpha_j + t_2 \beta_j)]^{-1/2} \right] \exp \left[- \left[r \frac{\mu_1^2}{\sigma_1^2} + (n-r) \frac{\mu_2^2}{\sigma_2^2} \right] / 2 + \sum_j \{ \mu_{1j}^2 / 2\sigma_{1j}^2 [1 - 2i(t_1 \alpha_j + t_2 \beta_j) \sigma_{1j}^2] \} \right]$$

In this case no simplification is afforded by the assumption of $\mu_1 = \mu_2$ or of $\sigma_1^2 = \sigma_2^2$, (for general values of α_j and β_j).

(b) Let $A=I$ and $B=J/n$. In this case we have the joint c.f. of $(\Sigma x_i^2, (\Sigma x_i)^2/n)$.

Now

$$(3.3) \quad |I - 2i(t_1 A + t_2 B) \Lambda_p| = \prod_{j=1}^n \{ 1 - 2it_1 \sigma_{1j}^2 \} |I - 2it_2 \frac{J}{n} \Lambda_p^*|$$

with

$$(3.4) \quad \Lambda_p^* = \Lambda_p (I - 2it_1 \Lambda_p)^{-1}.$$

Also

$$|I - 2it_2 (J/n) \Lambda_p^*| = (1 + \sum_{j=1}^n x_j)$$

where, for $j=1, 2, \dots, n$,

$$x_j = -2it_2 \sigma_{1j}^2 / n(1 - 2it_1 \sigma_{1j}^2).$$

Hence

$$(3.5) \quad |I - 2i(t_1 A + t_2 B) \Lambda_p| = (1 - 2it_1 \sigma_1^2)^r (1 - 2it_1 \sigma_2^2)^{n-r}$$

$$\left[1 - \frac{r}{n} \frac{2it_2 \sigma_1^2}{1 - 2it_1 \sigma_1^2} - \frac{n-r}{n} \frac{2it_2 \sigma_2^2}{1 - 2it_1 \sigma_2^2} \right]$$

In order to evaluate the quadratic expression in the exponent viz.,

$$(3.6) \quad \mu_p' \{ \Lambda_p - 2i \Lambda_p (t_1 A + t_2 B) \Lambda_p \}^{-1} \mu_p$$

we have to determine the inverse matrix involved. Now, in view of the nature of A and B,

$$\{ \Lambda_p - 2i \Lambda_p (t_1 A + t_2 B) \Lambda_p \}^{-1} = \Lambda_p^{-1} \{ T - 2it_2 J/n \}^{-1} \Lambda_p^{-1}$$

where

$$T = \Lambda_p^{-1} (I - 2it_1 \Lambda_p).$$

As before we refer to Rao (1965) and obtain

$$\{ T - (2it_2/n) J \}^{-1} = T^{-1} + \frac{2it_2}{n} \frac{T^{-1} J T^{-1}}{1 + (-2it_2/n) \underline{j}' T^{-1} \underline{j}}$$

and hence

$$(3.7) \quad \{ \Lambda_p - 2i \Lambda_p (t_1 A + t_2 B) \Lambda_p \}^{-1} = \Lambda_p^{-1} (I - 2it_1 \Lambda_p)^{-1}$$

$$+ \frac{2it_2}{n} \frac{(I - 2it_1 \Lambda_p)^{-1} J (I - 2it_1 \Lambda_p)^{-1}}{1 - (2it_2/n) \underline{j}' T^{-1} \underline{j}}$$

Upon applying (3.7) in (3.6) we get

$$\sum_{j=1}^n \left[\frac{\mu_{i_j}^2}{\sigma_{i_j}^2 (1 - 2it_1 \sigma_{i_j}^2)} \right] + \frac{2it_2}{n} \frac{\left[\sum_{j=1}^n \mu_{i_j} / (1 - 2it_1 \sigma_{i_j}^2) \right]^2}{\left[1 - \frac{2it_2}{n} \sum_{j=1}^n \{ \sigma_{i_j}^2 / (1 - 2it_1 \sigma_{i_j}^2) \} \right]}$$

By use of the identities defined earlier (2.7), we can simplify this further to obtain

$$(3.8) \quad \psi(t_1, t_2) = \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} \left\{ (1-2it_1\sigma_1^2)^{-r/2} (1-2it_1\sigma_2^2)^{-(n-r)/2} \right. \\ \left. \left[1 - (r/n) \frac{2it_2\sigma_1^2}{1-2it_1\sigma_1^2} - \{(n-r)/n\} \frac{2it_2\sigma_2^2}{1-2it_1\sigma_2^2} \right]^{1/2} \right\} \\ \exp \left\{ \left[2i(t_1+t_2) \left(\frac{r\mu_1^2}{1-2it_1\sigma_1^2} + \frac{(n-r)\mu_2^2}{1-2it_1\sigma_2^2} \right) - 2it_2 \frac{r(n-r)}{n} \right. \right. \\ \left. \left. \frac{(\mu_1 - \mu_2)^2}{(1-2it_1\sigma_1^2)(1-2it_1\sigma_2^2)} \right] / 2 \left[1 - \frac{2it_2}{n} \left(\frac{r\sigma_1^2}{1-2it_1\sigma_1^2} + \frac{(n-r)\sigma_2^2}{1-2it_1\sigma_2^2} \right) \right] \right\}$$

If $\sigma_1^2 = \sigma_2^2 = \lambda$ and $t_1 = -t_2$, then we obtain, from (3.8), the c.f. of $\Sigma(x_i - \bar{x})^2$ as presented in equation (2.6).

Also, if $\mu_1 = \mu_2 = \mu$ and $\sigma_1^2 = \sigma_2^2 = \lambda$, then we have in the normal case

$$(3.9) \quad (1-2it_1\lambda)^{-(n-1)/2} \{1-2i(t_1+t_2)\lambda\}^{-1/2} \exp[2i(t_1+t_2)\mu^2/2\{1-2i(t_1+t_2)\lambda\}].$$

(c) One way classification: Consider the between (treatments) and within sums of squares in one way ANOVA. In this case the quadratic forms are

$$T = \text{Between the sum of squares} = \underline{x}'[\Gamma/n - \tau/nk]\underline{x} \equiv \underline{x}'\underline{A}\underline{x}$$

$$W = \text{Within sum of squares} = \underline{x}'[I - \Gamma/n]\underline{x} \equiv \underline{x}'\underline{B}\underline{x}$$

with $\underline{x}' \equiv \{x_{ij}; i=1, \dots, k; j=1, 2, \dots, n\}$ a row vector of nk elements. Also τ is an $nk \times nk$ matrix of 1's everywhere. Γ is as defined earlier. Thus (with $N = nk$) if we assume $\sigma_1^2 = \sigma_2^2 = \lambda$, i.e., $\Lambda_p = \lambda I$, then

$$(3.10) \quad \psi_{T,W}(t_1, t_2) = \sum_{r=0}^N p^r (1-p)^{N-r} \sum_p \left\| 1 - 2it_1 \lambda \left(\frac{\Gamma}{n} - \frac{\tau}{nk} \right) - 2it_2 \lambda \left(1 - \frac{\Gamma}{n} \right) \right\|^{-1/2} \\ \exp \left[-\frac{1}{2} \underline{\mu}'_p \underline{\mu}_p + \frac{1}{2} \underline{\mu}'_p \{ \lambda I - 2i\lambda^2 (t_1 A + t_2 B) \}^{-1} \underline{\mu}_p \right].$$

The determinant in (3.10) can be seen to be (Subrahmaniam, 1968),

$$(1-2it_1\lambda)^{k-1}(1-2it_2\lambda)^{(N-k)}$$

Also we know that (e.g., Subrahmaniam, 1968)

$$\{\lambda I - 2i\lambda^2(t_1 A + t_2 B)\}^{-1} = (1/\lambda)H$$

where

$$H = \begin{pmatrix} U & V & V & \dots \\ V & U & V & \dots \\ V & V & U & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

with

$$u_{rr} = \frac{n-1}{n} \left[\frac{1}{(1-2it_2\lambda)} - \frac{1}{1-2it_1\lambda} \right] + \frac{1-2it_1\lambda/nk}{1-2it_1\lambda},$$

$$u_{rs} = \left[\frac{(2it_1\lambda/nk)}{1-2it_1\lambda} + \frac{1}{n} \left(\frac{1}{1-2it_2\lambda} - \frac{1}{1-2it_1\lambda} \right) \right], \quad r \neq s$$

and

$$V = \frac{(-2it_1\lambda/nk)}{(1-2it_1\lambda)} J.$$

Upon using these results in the exponent on the right hand side of (3.10), and substituting in (3.9), we have

$$(3.11) \quad \psi_{T,W}(t_1, t_2) = \sum_{r=0}^N p^r (1-p)^{N-r} \sum_p (1-2it_1\lambda)^{-(k-1)/2} (1-2it_2\lambda)^{-(N-k)/2}$$

$$\exp \left[\{it_2/(1-2it_2\lambda)\} \{r\mu_1^2 + (N-r)\mu_2^2\} \right.$$

$$\left. - \{it_1/(1-2it_1\lambda)\} \{r\mu_1 + \overline{N-r}\mu_2\}^2/N \right]$$

$$- \{(1-2it_2\lambda)^{-1} - (1-2it_1\lambda)^{-1}\} \left\{ \sum_j^k \mu_{ij}^2 / 2n\lambda \right\}].$$

If we put $\mu_1 = \mu_2$ we get the normal c.f.

$$(1-2it_1\lambda)^{-(k-1)/2}(1-2it_2\lambda)^{-(N-k)/2}.$$

Also, if $t_1 = t_2 = t$, we obtain the c.f. of the total sum of squares $\sum \sum (x_{ij} - \bar{x})^2$ to be

$$\psi_{\text{Total}}(t) = \sum_{r=0}^N p^r (1-p)^{N-r} \binom{N}{r} (1-2it\lambda)^{-(N-1)/2} \exp \left[\frac{2it\lambda}{1-2it\lambda} \delta_r^2 \right]$$

with $\delta_r^2 = r(N-r)(\mu_1 - \mu_2)^2 / 2N\lambda$. This agrees with the result in (2.6) for N observations.

Further, if $t_1 = 0$, the marginal c.f. (refer equation (2.11)) of the quadratic form $\sum \sum (x_{ij} - \bar{x}_i)^2$, i.e., the within sum of squares is obtained. A similar expression may be obtained for c.f. of T putting $t_2 = 0$ in (3.11).

4. Joint Distributions of Linear and Quadratic Forms

4.1 Let $\ell' \equiv (\ell_1, \ell_2, \dots, \ell_n)$ and $A \equiv \{\alpha_{ij}\}$ be a row vector and a matrix of known constants, respectively.

Consider their joint characteristic function

$$(4.1) \quad \psi(t_1, t_2) = E[\exp \{it_1 \ell' \underline{x} + it_2 \underline{x}' A \underline{x}\}].$$

We can write the exponent in (4.1) as (Subrahmaniam, 1966):

$$it_2 (\underline{x} - \underline{\nu})' A (\underline{x} - \underline{\nu}) - it_2 \underline{\nu}' A \underline{\nu}$$

with $\underline{\nu}$ satisfying the set of equations

$$(4.2) \quad \underline{\nu}' A = -(t_1/2t_2) \ell'.$$

Hence the c.f. $\psi(t_1, t_2)$ is seen to be

$$(4.3) \quad \psi(t_1, t_2) = \exp(-it_2 \underline{\nu}' A \underline{\nu}) E[\exp it_2 (\underline{x} - \underline{\nu})' A (\underline{x} - \underline{\nu})].$$

Now, recognising that $\phi_p^{(r)}$ is a multinormal p.d.f., we have,

$$(4.4) \quad \psi(t_1, t_2) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p |I - 2it_2 A \Lambda_p|^{-1/2} \exp \left[-\frac{1}{2} (\underline{\mu}_p - \underline{\nu})' \Lambda_p^{-1} (\underline{\mu}_p - \underline{\nu}) \right. \\ \left. + \frac{1}{2} (\underline{\mu}_p - \underline{\nu})' \{ \Lambda_p - 2it_2 \Lambda_p A \Lambda_p \}^{-1} (\underline{\mu}_p - \underline{\nu}) - it_2 \underline{\nu}' A \underline{\nu} \right].$$

An alternative form of expression in the exponent involving only $\underline{\ell}$ and A is more useful. This may be obtained as follows:

Rearranging the term in (4.4) in powers of $\underline{v}$ we have

$$\begin{aligned} & -it_2(\underline{v}'A\underline{v}) + \frac{1}{2} \underline{v}' \{ \Lambda_p - 2it_2 \Lambda_p A \Lambda_p \}^{-1} \underline{v} - \frac{1}{2} \underline{v}' \Lambda_p^{-1} \underline{v} - \underline{v}' \{ \Lambda_p - 2it_2 \Lambda_p A \Lambda_p \}^{-1} \underline{\mu}_p \\ & + \underline{v}' \Lambda_p^{-1} \underline{\mu}_p + \frac{1}{2} \underline{\mu}_p' \{ \Lambda_p - 2it_2 \Lambda_p A \Lambda_p \}^{-1} \underline{\mu}_p - \frac{1}{2} \underline{\mu}_p' \Lambda_p^{-1} \underline{\mu}_p \end{aligned}$$

Hence

$$\begin{aligned} (4.5) \quad \psi(t_1, t_2) &= \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p |I - 2it_2 A \Lambda_p|^{-1/2} \exp[-(\frac{1}{2}) t_1^2 \underline{\ell}' \{ \Lambda_p - 2it_2 A \}^{-1} \underline{\ell} \\ &+ it_1 \underline{\ell}' \{ I - 2it_2 \Lambda_p A \}^{-1} \underline{\mu}_p + it_2 \underline{\mu}_p' \{ I - 2it_2 A \Lambda_p \}^{-1} A \underline{\mu}_p]. \end{aligned}$$

4.2 Special cases:

(a) $A = I$ and $\underline{\ell} = \underline{j}$; i.e., $(\Sigma x_i, \Sigma x_i^2)$: In this case, simplifying the various expressions, using (2.7), we have

$$\begin{aligned} \psi(t_1, t_2) &= \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} (1-2it_2 \sigma_1^2)^{-r/2} (1-2it_2 \sigma_2^2)^{-(n-r)/2} \exp[-(t_1^2/2) \\ &\{ r \sigma_1^2 / (1-2it_2 \sigma_1^2) + (n-r) \sigma_2^2 / (1-2it_2 \sigma_2^2) \} + it_1 \{ r \mu_1 / (1-2it_2 \sigma_1^2) + \\ &(n-r) \mu_2 / (1-2it_2 \sigma_2^2) \} + it_2 \{ r \mu_1^2 / (1-2it_2 \sigma_1^2) + (n-r) \mu_2^2 / (1-2it_2 \sigma_2^2) \}]. \end{aligned}$$

Some simplifications are afforded if we make assumptions about the parameters involved. If $\mu_1 = \mu_2 = 0$, then

$$\begin{aligned} \psi(t_1, t_2) &= [\{ p \exp \{ -t_1^2 \sigma_1^2 / 2(1-2it_2 \sigma_1^2) \} / (1-2it_2 \sigma_1^2)^{1/2} \\ &+ (1-p) \exp \{ -t_1^2 \sigma_2^2 / 2(1-2it_2 \sigma_2^2) \} / (1-2it_2 \sigma_2^2)^{1/2} \}]^n \end{aligned}$$

If $\sigma_1^2 = \sigma_2^2 = \lambda$, then

$$\begin{aligned} \psi(t_1, t_2) &= (1-2it_2 \lambda)^{-n/2} \exp \{ -n \lambda t_1^2 / 2(1-2it_2 \lambda) \} \\ &\sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} \exp \{ (it_1 C_1 + it_2 C_2) / (1-2it_2 \lambda) \} \end{aligned}$$

with, for $j = 1, 2$,

$$C_j = r\mu_1^j + (n-r)\mu_2^j$$

which corresponds with (6) of Hyrenius (1949). It should be noted that his equation seems to have a few errors.

(b) If $A = (I - J/n)$, $\underline{\ell}' = \underline{j}'$, then we have the joint characteristic function of $(\Sigma x_i, \Sigma(x_i - \bar{x})^2)$. Here

$$(4.7) \quad |I - 2it_2 A \Lambda_p|^{-1/2} = |I - 2it_2 \Lambda_p|^{-1/2} |I - \frac{2it_2}{n} J \Lambda_p^*|^{-1/2}$$

with

$$\Lambda_p^* = \Lambda_p (I - 2it_2 \Lambda_p)^{-1}.$$

From section 3.2 (b) we know that the expression in (4.7) is

$$(1 - 2it_2 \sigma_1^2)^{-r/2} (1 - 2it_2 \sigma_2^2)^{-(n-r)/2} \left\{ 1 + \frac{2it_2}{n} \left[\frac{r\sigma_1^2}{1 - 2it_2 \sigma_1^2} + \frac{(n-r)\sigma_2^2}{1 - 2it_2 \sigma_2^2} \right] \right\}^{-1/2}$$

Also

$$(4.8) \quad \{\Lambda_p^{-1} - 2it_2 A\}^{-1} = \{\Lambda_p^{-1} - 2it_2 I\}^{-1} - \frac{2it_2}{n} \frac{(\Lambda_p^{-1} - 2it_2 I)^{-1} J (\Lambda_p^{-1} - 2it_2 I)^{-1}}{1 + (2it_2/n) \underline{j}' (\Lambda^{-1} - 2it_2 I)^{-1} \underline{j}}$$

But the matrix

$$\{\Lambda_p^{-1} - 2it_2 I\}^{-1} = \text{diag} \left\{ \frac{\sigma_{i_1}^2}{1 - 2it_2 \sigma_{i_1}^2}, \dots, \frac{\sigma_{i_n}^2}{1 - 2it_2 \sigma_{i_n}^2} \right\}$$

and, therefore,

$$\underline{\ell}' (\Lambda_p^{-1} - 2it_2 A)^{-1} \underline{\ell} = \left[\frac{r\sigma_1^2}{(1 - 2it_2 \sigma_1^2)} + \frac{(n-r)\sigma_2^2}{(1 - 2it_2 \sigma_2^2)} \right] / \left[\frac{r}{n(1 - 2it_2 \sigma_1^2)} + \frac{(n-1)}{n(1 - 2it_2 \sigma_2^2)} \right],$$

$$\mu_p' \{I - 2it_2 A_p A\}^{-1} \mu_p = \frac{[r\mu_1 / (1 - 2it_2 \sigma_1^2) + (n-r)\mu_2 / (1 - 2it_2 \sigma_2^2)] / [r/n(1 - 2it_2 \sigma_1^2) + (n-r)/n(1 - 2it_2 \sigma_2^2)]}{1}$$

and

$$\mu_p' \{I - 2it_2 A_p A\}^{-1} A \mu_p = \frac{r(n-r) \{(\mu_1 - \mu_2)^2 - \mu_1 \mu_2 (\sigma_1^{-2} - \sigma_2^{-2})^2\}}{r(1 - 2it_2 \sigma_1^2) + (n-r)(1 - 2it_2 \sigma_1^2)}$$

Thus the joint characteristic function is

$$(4.9) \quad \psi(t_1, t_2) = \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} (1 - 2it_2 \sigma_1^2)^{-r/2} (1 - 2it_2 \sigma_2^2)^{-(n-r)/2} \left[\frac{r}{n(1 - 2it_2 \sigma_1^2)} + \frac{(n-r)}{n(1 - 2it_2 \sigma_2^2)} \right]^{-1/2} \exp \left\{ \left[\frac{-t_1^2}{2} [r\sigma_1^2 / (1 - 2it_2 \sigma_1^2) + (n-r)\sigma_2^2 / (1 - 2it_2 \sigma_2^2)] + it_1 [r\mu_1 / (1 - 2it_2 \sigma_1^2) + (n-r)\mu_2 / (1 - 2it_2 \sigma_2^2)] + it_2 [(r(n-r)/n) \{(\mu_1 - \mu_2)^2 - \mu_1 \mu_2 (\sigma_1^{-2} - \sigma_2^{-2})^2\} / (1 - 2it_2 \sigma_1^2)(1 - 2it_2 \sigma_2^2)] \right] / \{[r/n(1 - 2it_2 \sigma_1^2)] + [(n-r)/n(1 - 2it_2 \sigma_2^2)]\} \right\}$$

If we put $t_1 = 0$ in (4.9), we have the marginal c.f. of $\Sigma(x_1 - \bar{x})^2$:

$$(4.10) \quad \psi(0, t_2) = \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} (1 - 2it_2 \sigma_1^2)^{-r/2} (1 - 2it_2 \sigma_2^2)^{-(n-r)/2} \left[\{r/n(1 - 2it_2 \sigma_1^2)\} + \{(n-r)/n(1 - 2it_2 \sigma_2^2)\} \right]^{-1/2} \exp \left[it_2 \{r(n-r)/n\} \{(\mu_1 - \mu_2)^2 - \mu_1 \mu_2 (\sigma_1^{-2} - \sigma_2^{-2})^2\} / \{[r/n(1 - 2it_2 \sigma_1^2)] + [(n-r)/n(1 - 2it_2 \sigma_2^2)]\} \right].$$

The c.f. (4.10) generalises the c.f. obtained earlier in (2.6) under assumption

$$\sigma_1^2 = \sigma_2^2 = \lambda.$$

If $\mu_1 = \mu_2 = 0$ in (4.9), then

$$\begin{aligned}
 (4.11) \quad \psi(t_1, t_2) &= \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} (1-2it_2\sigma_1^2)^{-r/2} (1-2it_2\sigma_2^2)^{-(n-r)/2} \\
 &\quad \{ [r/n(1-2it_2\sigma_1^2)] + [(n-r)/n(1-2it_2\sigma_2^2)] \}^{-1/2} \\
 &\quad \exp \{ -it_1^2 \{ [r\sigma_1^2/(1-2it_2\sigma_1^2)] + [(n-r)\sigma_2^2/(1-2it_2\sigma_2^2)] \} / \\
 &\quad 2 \{ [r/n(1-2it_2\sigma_1^2)] + [(n-r)/n(1-2it_2\sigma_2^2)] \} \}
 \end{aligned}$$

On the other hand, if $\sigma_1^2 = \sigma_2^2 = \lambda$, then

$$\begin{aligned}
 (4.12) \quad \psi(t_1, t_2) &= \sum_{r=0}^n \binom{n}{r} p^r (1-p)^{n-r} (1-2it_2\lambda)^{-(n-1)/2} \\
 &\quad \exp \{ -n\lambda t_1^2/2 + it_1 \{ r\mu_1 + (n-r)\mu_2 \} + it_2 \{ r(n-r)/n \} (\mu_1 - \mu_2)^2 / (1-2it_2\lambda) \}
 \end{aligned}$$

(c) Joint distribution of the regression coefficient and the residual sum of squares (conditional):

Let

$$\begin{aligned}
 b &= \Sigma(y_i - \bar{y})x_i / \sqrt{\Sigma(y_i - \bar{y})^2} \\
 s^2 &= \Sigma(x_i - \bar{x})^2 - \{ \Sigma(y_i - \bar{y})x_i \}^2 / \Sigma(y_i - \bar{y})^2
 \end{aligned}$$

and hence in the matrix form let

$$\underline{\ell}' = \{ (y_1 - \bar{y})/c, (y_2 - \bar{y})/c, \dots, (y_n - \bar{y})/c \}$$

$$A = \{ I - J/n - Q \}$$

where

$$Q = \underline{\ell} \underline{\ell}'$$

and

$$c = [\Sigma(y_i - \bar{y})^2]^{1/2}.$$

The joint c.f. of $\mathbf{b}$ and s^2 is

$$(4.13) \quad \psi(t_1, t_2) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum \{I - 2it_2 \Lambda_p + 2it_2 J \Lambda_p / n + 2it_2 Q \Lambda_p\}^{-1/2} \exp[-t_1^2 \underline{\mu}' \{\Lambda_p^{-1} - 2it_2(I - J/n - Q)\}^{-1} \underline{\mu} / 2 + it_1 \underline{\mu}' \{I - 2it_2 \Lambda_p(I - J/n - Q)\}^{-1} \underline{\mu}_p + it_2 \underline{\mu}' \{I - 2it_2(I - J/n - Q) \Lambda_p\}^{-1} (I - J/n - Q) \underline{\mu}_p\}].$$

Let us assume here that $\sigma_1^2 = \sigma_2^2 = \lambda$. Then the determinant in (4.13) can be shown to be equal to (Subrahmaniam, 1968)

$$(1 - 2it_2 \lambda)^{(n-2)}.$$

Also, the various matrices involved in the exponent can be determined by reference to the same paper.

The c.f. of $(\mathbf{b}, s^2)$ is thus

$$(4.14) \quad \psi(t_1, t_2) = \sum_{r=0}^n p^r (1-p)^{n-r} \sum_p (1 - 2it_2 \lambda)^{-(n-2)/2} [\exp\{-\lambda t_1^2 / 2 + it_1 \underline{\mu}' \underline{\mu}_p + [it_2 / (1 - 2it_2 \lambda)] [\underline{\mu}' \underline{\mu}_p - \underline{\mu}'_p J \underline{\mu}_p / n - \underline{\mu}'_p Q \underline{\mu}_p]\}]$$

If $\mu_1 = \mu_2$, we have the usual expression

$$\psi(t_1, t_2) = (1 - 2it_2 \lambda)^{-(n-2)/2} \exp(-\lambda t_1^2 / 2).$$

Since the quantity $\underline{\mu}' \underline{\mu}_p$ is not symmetric with respect to the permutations in the vector $(\mu_1, \dots, \mu_n)$ we do not have any simple closed expression for $\sum_p$ in (4.14).

15. Discussion

This paper presents the characteristic functions of various statis-

tics one encounters in inference. In some cases we have also obtained the distribution functions. The results should be of great interest in studying the effects of departure from normality on the standard tests of significance. It appears that no work has been done on this problem using this model. For example, let us consider the one way classification:

$$Y_{ij} = \alpha + \beta_i + \epsilon_{ij} \quad i = 1, 2, \dots, k; j = 1, \dots, n,$$

where ϵ_{ij} are i.i.d. random variables having the p.d.f. (1.1). This representation gives the model for Y_{ij} as

$$E(Y_{ij}) = \alpha + \beta_i + p\mu_1 + (1-p)\mu_2.$$

Hence the treatment means are estimated as usual. The effect of non-normality is felt in the level of significance in view of the fact that the distribution of the ratio T/W is no longer the familiar F ratio even under H_0 . A study of the effects of this type of non-normality are to be undertaken.

Acknowledgements

The author wishes to thank an associate editor and Professor J.H. Venter for their help in condensing an original version of the paper.

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